



Dr. A. Lakshmanan
CEO & MD - IFFCO Nanoventions



Mr. Yogendra Kumar
Director, Marketing - IFFCO

interestingly it has triggered a cascading effect with many companies venturing into this vertical. This clearly underscores the foresight of Dr. Awasthi in introducing this revolutionary concept and potential of nanofertiliser technology in the fertiliser industry.

Dr. Awasthi's keen interest and resolute faith in nanotechnology is remarkable, which has encouraged IFFCO to embark on the process of introducing more nanoformulations in agriculture including granular nanofertiliser for soil (root zone) application. The concept is to promote foliar nutrition through liquid nanofertilisers and soil mediated nutrition through granular nanofertiliser. This would certainly revolutionize the crop nutrition and more importantly transform agricultural methodologies towards carbon neutral farming.

By adopting this cutting-edge nano technology early on, Dr. Awasthi and

his team have introduced innovative products such as:

➤ IFFCO Nano Urea: A liquid nano fertiliser that provides nitrogen to plants precisely.

➤ IFFCO Nano DAP (Diammonium Phosphate): A nano-based phosphorus fertiliser that enhances phosphorus availability to plants.

➤ Nano Zinc and Nano Copper: The trace element formulations crafted as nanoclusters with highest use efficiency and ensure bioavailability of zinc and copper through fertiliser-crop-human continuum.

These innovative nano products from IFFCO are helping curtail fertiliser usage by up to 50%, minimising Green House Gas (GHG) emission in particular the nitrous oxide and environmental pollution by fertiliser run off and ultimately improve crop yield and quality. This is a major step towards an 'Ever green

revolution' that is truly green!

"Nanofertiliser would not have witnessed commercial success without the mammoth contribution by Dr. Awasthi," says Dr. A. Lakshmanan, the Managing Director of IFFCO-Nanoventions, who is leading the nano research for IFFCO and owns many Patents in nanofertiliser. "Introducing commercial nano formulations in fertiliser industry needs lot of investment, dedicated research, biosafety and bio efficacy studies and large scale field validation and most of all the constant guidance and motivation from the decision makers. Dr. Awasthi understood these prerequisites and offered the same to me and our team that led to the success. As nanofertiliser is a fast evolving field and IFFCO, being the pioneer in this sector, had to face lot of challenges, but Dr. Awasthi has been rock solid in his aspiration of introducing nanofertilisers in agriculture. Having pioneered this Nano Revolution in the global agriculture, it would be apt to call Dr. Awasthi as "Father of Nanotechnology in Agriculture", opines Dr. Lakshmanan.

Under the guidance of Dr. Awasthi, IFFCO has launched a "Drone as a Service" (DaaS) scheme to spray nano fertiliser and other products on all crops across India. This fertiliser behemoth of India has trained over 1500 rural entrepreneurs across the country to operate Agri drones. In an ongoing process, A Memorandum of Agreement (MoA) has already been signed with leading Agri Drone Service providers to successfully implement this project that would help the farmers in the effective and safe delivery of agro chemicals in farming for reaping better crops. In his message for the IFFCO family, cooperatives, associates, and partners while highlighting the performance of the cooperative in FY 2024-2025, Dr. Awasthi said, "As a part of PM's initiative of Namo Drone Didi, this year IFFCO provided drone training for spraying liquid fertiliser to 300 women farmers as Namo Drone Didi. By breaking barriers and embracing technology Namo Drone Didi's are showcasing the strength and potential of rural women. We also trained more than 1764 rural drone entrepreneurs thus promoting an ecosystem focussed on self-sufficiency and sustainability."

In the same IFFCO FY 24-25 message,

